

ENVARIABELSANALYS DEL 2

SEMINARIUM 6: TRIGONOMETRISKA UTTRYCK

$$\int f(\sin(x)) \cdot \cos(x) dx = \left[\begin{array}{l} \sin(x) = t \\ \cos(x) dx = dt \end{array} \right]$$

$$= \int f(t) dt$$

ELLER

$$\int f(\cos(x)) \cdot \sin(x) dx = \left[\begin{array}{l} \cos(x) = t \\ \sin(x) dx = -dt \\ \sin(x) dx = -dt \end{array} \right]$$

$$= -\int f(t) dt$$

EX: $\int \sin^5(x) dx = \int \underbrace{\sin^4(x)}_{f(\sin(x))} \cdot \sin(x) dx$

= $\left[\begin{array}{l} \text{Trigonometrisk identitet} \\ \sin^2(x) = 1 - \cos^2(x) \end{array} \right] = \int \frac{(1 - \cos^2(x))^2}{f(\cos(x))} \sin(x) dx$

= $\left[\begin{array}{l} \cos(x) = t \\ -\sin(x) dx = dt \\ \sin(x) dx = -dt \end{array} \right] = \int (1 - t^2)^2 (-dt)$

$$= -\int (1 - 2t^2 + t^4) dt$$

$$= -\left(t - 2 \frac{t^3}{3} + \frac{t^5}{5} \right) + C =$$

$$= -\cos(x) + \frac{2}{3} \cos^3(x) + \frac{1}{5} \cos^5(x) + C$$

SVAR:

$f(x) =$ 

$$\text{Ex. } \int \sin^2(x) dx = \left[\begin{array}{l} \text{jämma} \\ \text{exponenter} \end{array} \right]$$

- [$\cos(2x)$ är av intresse då här
vi jämma exponenter]

$$= \left[\begin{array}{l} \cos(2x) = \cos^2(x) - \sin^2(x) \\ \cos(2x) = ((1 - \sin^2(x)) - \sin^2(x)) \\ \Leftrightarrow \cos(2x) = 1 - 2\sin^2(x) \\ \cos 2x = \cos^2(x) - (1 - \cos^2(x)) \\ \Leftrightarrow \cos(2x) = 2\cos^2(x) - 1 \end{array} \right]$$

$$= \left[\begin{array}{l} \cos(2x) = 1 - 2\sin^2(x) \\ \Leftrightarrow \sin^2(x) = \frac{1 - \cos(2x)}{2} \end{array} \right]$$

$$= \int \left(\frac{1}{2} (1 - \cos(2x)) \right) dx = \int \left(\frac{1}{2} - \frac{1}{2} \cos(2x) \right) dx$$

$$= \frac{1}{2}x - \frac{1}{2} \frac{\sin(2x)}{2} + C$$

$$= \frac{1}{2}x - \frac{1}{4} \sin(2x) + C$$

ALTERNATIV LÖSNING (ej viktig)

$$= \int \frac{1}{2} dx - \frac{1}{2} \int \cos(2x) dx =$$

$$= \left[\begin{array}{l} 2x = t \\ 2dx = dt \\ dx = \frac{1}{2} dt \end{array} \right] = \int \frac{1}{2} dx - \frac{1}{2} \int \cos(t) \cdot \frac{1}{2} dt$$

$$= \int \frac{1}{2} dt - \frac{1}{4} \int \cos(t) dt =$$

$$= \frac{1}{2}x - \frac{1}{4} \sin(t) + C$$

$$f(x) = \frac{1}{2}x - \frac{1}{4} \sin(2x) + C$$

Allmänt

Om vi har

$\int \sin^n(x) dx$ eller $\int \cos^n(x) dx$, n är

heltal, $n \geq 1$

• om n är udda: utnyttja trigonometriska

• om n är jämn: Då gäller formeln
för dubbelvinklar

$$- \cos(2x)$$

$$- \sin(2x)$$

Ex: $\int \frac{\sin(x) \cos(x)}{4 - \cos^2(x)} dx =$

$$= \int \frac{\cos(x)}{4 - \cos^2(x)} \cdot \sin(x) dx = \left[\begin{array}{l} \cos(x) = t \\ -\sin(x) dx = dt \end{array} \right]$$

$$= \int \frac{t}{4 - t^2} (-dt) = \int \frac{-t}{4 - t^2} dt \quad \left[\sin(x) dx = -dt \right]$$

$$= \int \left(\frac{1}{2} \cdot \frac{-2t}{4 - t^2} \right) dt = \frac{1}{2} \int \frac{-2t}{4 - t^2} dt$$

$$= \frac{1}{2} \ln|4 - t^2| + C$$

SVAR:

$$F(x) = \frac{1}{2} \ln|4 - t^2| + C = \frac{1}{2} \ln|4 - \cos^2 t| + C$$

ALTERNATIV LÖSNING

$$\left[\begin{array}{l} 4 - \cos^2(x) = t \\ -2 \cos(x) \sin(x) dx = dt \\ \cos(x) \sin(x) dx = -\frac{1}{2} dt \end{array} \right] = \int \frac{\sin(x) \cos(x)}{4 - \cos^2(x)} dx$$

$$= \int \frac{1}{t} \cdot \frac{1}{2} dt$$

$$= \frac{1}{2} \int \frac{1}{t} dt = \frac{1}{2} \ln|t| + C \Rightarrow \frac{1}{2} \ln|4 - \cos^2(x)| + C$$

$$\underline{\text{Ex:}} \int \frac{1}{\sin(x)} dx = \left[\begin{array}{l} \tan\left(\frac{x}{2}\right) = t \\ \frac{x}{2} = \arctan(t) \\ x = 2 \arctan(t) \\ dx = 2 \cdot \frac{1}{1+t^2} dt \end{array} \right]$$

$$\begin{aligned} &= \left[\begin{array}{l} \sin(x) = \sin\left(2 \cdot \frac{x}{2}\right) = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) \\ = \frac{2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)}{1} = \frac{2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)}{\sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right)} \\ = \frac{2 \sin\left(\frac{x}{2}\right) \cancel{\cos\left(\frac{x}{2}\right)}}{\cos^2\left(\frac{x}{2}\right)} = \frac{2 \tan\left(\frac{x}{2}\right)}{\tan^2\left(\frac{x}{2}\right) + 1} = \frac{2t}{t^2+1} \\ \frac{\sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right)}{\cos^2\left(\frac{x}{2}\right) \cos^2\left(\frac{x}{2}\right)} \end{array} \right] \end{aligned}$$

$$= \int \frac{1}{\frac{2t}{t^2+1}} \cdot \frac{2}{t^2+1} dt = \int \frac{1}{t} dt = \ln|t| + C$$

$$= \ln\left|\tan\left(\frac{x}{2}\right)\right| + C$$

SVAR:

$$F(x) = \ln\left(\tan\left(\frac{x}{2}\right)\right) + C$$

$$\int \frac{1}{\sin(x)} dx = \int \frac{\sin(x)}{\sin^2(x)} dx = \left[\begin{array}{l} \text{trigonometrisch} \\ \text{erweitern} \end{array} \right]$$

$$= \int \frac{\sin(x)}{1 - \cos^2(x)} dx = \left[\begin{array}{l} \cos(x) = t \\ -\sin(x) dx = dt \\ \sin(x) dx = -dt \end{array} \right]$$

$$= \int \frac{-dt}{1-t^2} = \int \frac{1}{t^2-1} dt = \int \frac{1}{(t-1)(t+1)} dt$$

$$= \left[\begin{array}{l} \text{PARTIELLE BRUCHZERLEGUNG} \\ \frac{1}{(t-1)(t+1)} = \frac{A}{(t-1)} + \frac{B}{(t+1)} \\ A = \frac{1}{2} \\ B = -\frac{1}{2} \end{array} \right] =$$

$$= \int \frac{1}{2} \cdot \frac{1}{(t-1)} dt + \int \left(-\frac{1}{2}\right) \cdot \frac{1}{t+1} dt$$

$$= \frac{1}{2} \int \frac{1}{t-1} dt - \frac{1}{2} \int \frac{1}{t+1} dt =$$

$$= -\frac{1}{2} \ln|t-1| - \frac{1}{2} \ln|t+1| + C$$

$$= -\frac{1}{2} \ln|\cos(x)-1| - \frac{1}{2} \ln(\cos(x)+1) + C$$

SVAR
 $\int \frac{1}{\sin(x)} dx =$ 

ROTUTRUCK

S. 267

Ex:

$$\int \frac{x^3}{\sqrt{x^2+1}} dx = \int \frac{x^2 \cdot x}{\sqrt{x^2+1}} dx$$

$$= \left[\begin{array}{l} x^2 = t \\ 2x dx = dt \\ x dx = \frac{1}{2} dt \end{array} \right] = \int \frac{t}{\sqrt{t+1}} \cdot \frac{1}{2} dt$$

$$= \frac{1}{2} \int \frac{t}{\sqrt{t+1}} dt = \left[\begin{array}{l} \sqrt{t+1} = u \\ u^2 = t+1 \\ dt = 2u du \\ t = u^2 - 1 \end{array} \right]$$

$$= \frac{1}{2} \int \frac{u^2 - 1}{u} \cdot 2u du$$

$$= \frac{1}{2} \cdot 2 \int (u^2 - 1) du = \int u^2 du - \int 1 du$$

$$= \frac{u^3}{3} - \frac{u^2}{2} \quad | \quad u = \sqrt{t+1}$$

$$= \frac{(\sqrt{t+1})^3}{3} - \frac{(\sqrt{t+1})^2}{2} \quad | \quad t = x^2$$

$$= \frac{(\sqrt{x^2+1})^3}{3} - \frac{(\sqrt{x^2+1})^2}{2}$$

$$= \frac{((x^2+1)^{\frac{1}{2}})^3}{3} - \frac{x^2+1}{2}$$

$$= \frac{(x^2+1)^{\frac{3}{2}}}{3} - \frac{x^2+1}{2} = \frac{(x^2+1)^{\frac{1}{2} + \frac{2}{2}}}{3} - \frac{x^2+1}{2}$$

$$= \frac{\sqrt{x^2+1}(x^2+1)}{3} - \frac{x^2+1}{2} + C$$

$$\text{Ex } \int \sqrt{x^2+1} dx = \left[\begin{array}{l} \text{tabel s. 267} \\ x = \tan(t) \\ -\frac{\pi}{2} < t < \frac{\pi}{2} \end{array} \right]$$

$$x = \tan t$$

$$dx = \frac{1}{\cos^2(t)} dt$$

$$\tan(t) = \frac{\sin(t)}{\cos(t)}, \quad t = \arctan(x)$$

$$(\tan(t))'$$

$$= \frac{\cos(t) \cos(t) + \sin(t) \sin(t)}{\cos^2(t)} = \frac{\cos^2(t) + \sin^2(t)}{\cos^2(t)}$$

$$= \frac{1}{\cos^2(t)}$$

$$= \int (\sqrt{\tan^2(t) + 1} \cdot \frac{1}{\cos^2(t)}) dt$$

$$= \int \sqrt{\frac{\sin^2(t)}{\cos^2(t)} + \frac{\cos^2(t)}{\cos^2(t)}} \cdot \frac{1}{\cos^2(t)} dt$$

$$= \int \sqrt{\frac{\sin^2(t) + \cos^2(t)}{\cos^2(t)}} \cdot \frac{1}{\cos^2(t)} dt$$

$$= \left[\begin{array}{l} \text{Trigonometrische} \\ \text{relation} \end{array} \right] = \int \sqrt{\frac{1}{\cos^2(t)}} \cdot \frac{1}{\cos^2(t)} dt$$

$$= \int \frac{1}{|\cos(t)|} \cdot \frac{1}{\cos^2(t)} dt = \left[\begin{array}{l} -\frac{\pi}{2} < t < \frac{\pi}{2} \\ \cos t > 0 \\ 1 \cos t = \cos t \end{array} \right]$$

$$= \int \frac{1}{\cos(t)} \cdot \frac{1}{\cos^2(t)} dt$$

$$= \int \frac{1}{\cos^3(t)} dt = \left[t = \tan\left(\frac{u}{2}\right) \right]$$

Ex:

$$\int \sqrt{x+3} dx = \left[\begin{array}{l} \sqrt{x+3} = t \\ x+3 = t^2 \\ dx = 2t dt \end{array} \right]$$

$$= \int t \cdot 2t dt = \int 2t^2 dt = 2 \int t^2 dt$$

$$= 2 \frac{t^3}{3} + C$$

$$\frac{2}{3} (\sqrt{x+3})^3 + C$$

$$= \frac{2}{3} ((x+3)^{\frac{1}{2}})^{\frac{6}{2}} = \frac{2}{3} (x+3)^{\frac{3}{2}} =$$

$$= \frac{2}{3} (x+3)^{\frac{1}{2} + \frac{2}{2}} = \frac{2}{3} (x+3) \sqrt{x+3} + C$$

SVAR:

$$F(x) = \sqrt{x+3} (x+3) + C$$